Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

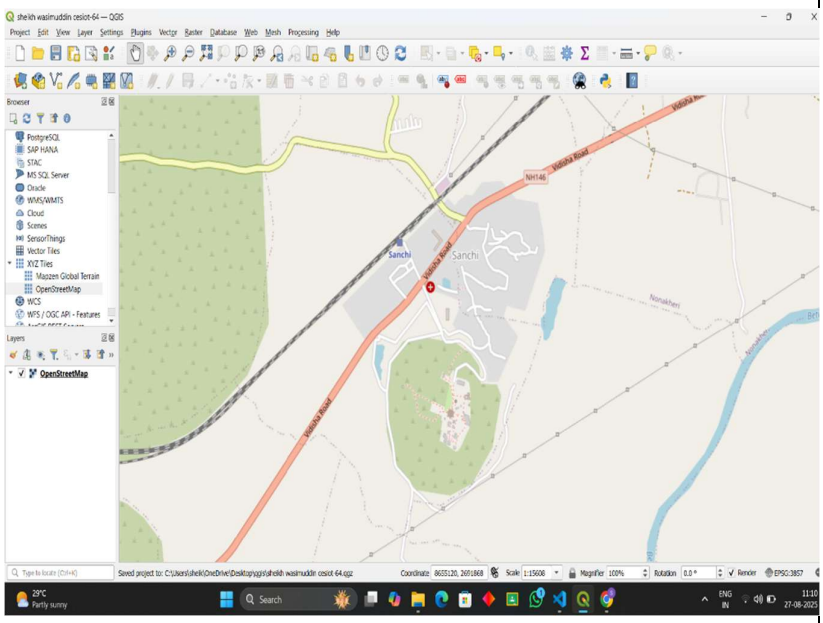
"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

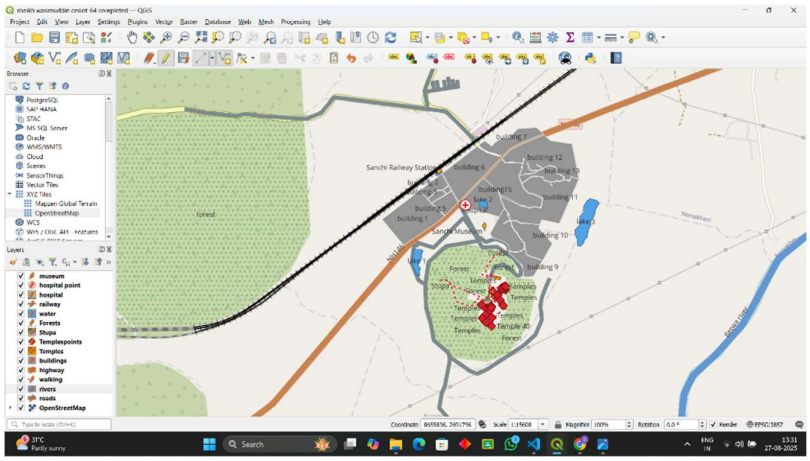
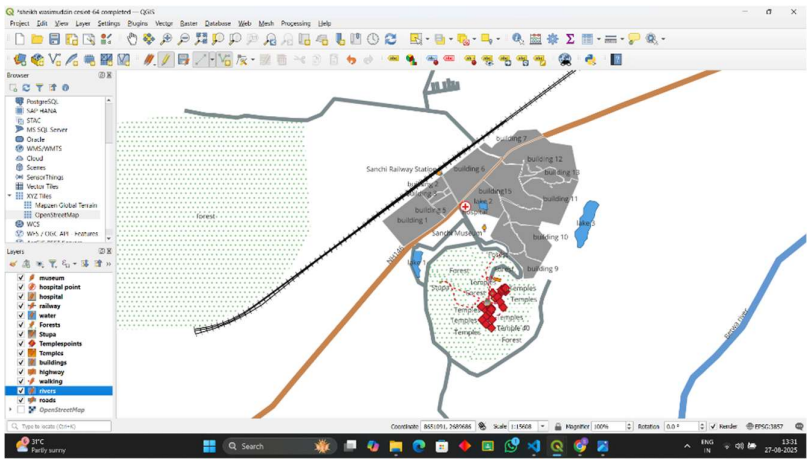
Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**04**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Produce a publication-quality map from vector layers, using Data View and Layout view.
Problem Definition	The task involves producing a publication-quality map from vector layers by using QGIS's Data View and Layout View to include appropriate cartographic elements such as legends, scale bars, and labels.
Theory (100 words)	Creating a professional-quality map generally starts in the **data preparation stage** within GIS software such as QGIS or ArcGIS. The first step is to load the necessary spatial datasets and confirm that all layers share a consistent **Coordinate Reference System (CRS)** so they align correctly. Once the data is loaded, the next task is **layer styling and arrangement**. This includes assigning appropriate colors, line types, and fill patterns, as well as categorizing data for better interpretation. The drawing order of layers must also be carefully managed so that important features remain visible. Afterward, **labels are applied** to features based on their attribute information. Adjustments in font size, placement, and style are made to improve readability and to minimize overlaps. The process then moves into the **map layout stage**, where the cartographic design is finalized. A layout is created, the map extent is defined, and a map frame is inserted from the data view. Supporting map elements such as a legend, north arrow, scale bar, and title are added, along with annotations if necessary. These elements should be arranged systematically to avoid visual clutter. Finally, the map undergoes **refinement** to ensure high quality. This involves checking the overall balance of colors, verifying that the scale is correct, and including additional details such as credits, author name, and data source information.

Procedure and Execution (100 Words)	Implementation Steps: ➢ Open QGIS → Blank map canvas appears. ➢ Open the Openstreet Map from XYZ Tiles. ➢ Add Line Shapefile for Roads. ➢ Add Polygon Shapefile for grassField. ➢ Add Polygon Shapefile for Building. ➢ Add Polygon Shapefile for WaterField. ➢ Add Line Shapefile for rivers. ➢ Add Point Shapefile for Temples. ➢ Add line Shapefile for Railway. Stepwise Screenshots with steps: 1) Open the Openstreet Map from XYZ Tiles. Now search for Locations like I Have done it for Sanchi (M.P). 

	2) Now Create Layer of polygons , Lines , Point.  
Output Analysis	By following the cartographic process in GIS, a well-structured and publication-quality map is generated. The proper use of CRS ensures that all spatial layers are accurately aligned, while effective symbolization and classification enhance the visual distinction between different features. Labeling improves interpretability by clearly identifying map elements without causing clutter. In the layout stage, the addition of essential components such as the legend, north arrow, scale bar, and title gives the map a

	professional appearance and helps users understand spatial information easily. Final refinements in design, such as colour balance and readability checks, ensure that the output communicates information clearly and aesthetically. Thus, the final map serves as both an analytical and presentation tool, meeting professional cartographic standards.
Conclusion	The practical was successfully completed, resulting in the creation of a clear and publication-quality map. All layers were properly aligned through CRS verification, symbolized for better visualization, and labeled for clarity. In the layout view, map elements such as title, legend, scale bar, and north arrow were neatly arranged, and final adjustments improved readability. This demonstrates the importance of cartographic design in effectively presenting spatial data.
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